

DDA5003: Stochastic Processes

Assignment 1

Question 1. Conditional probability

A machine is either working (W) or faulty (F). Suppose $\mathbb{P}(W) = 3/4$ and $\mathbb{P}(F) = 1/4$. An alarm sounds with probability $1/5$ when the machine is working and with probability $4/5$ when it is faulty. Let A be the event that the alarm sounds.

- (a) Find $\mathbb{P}(A)$.
- (b) Given that the alarm sounds, find $\mathbb{P}(F \mid A)$.

Question 2. Transition matrices and the Markov property

Consider the matrices

$$A = \begin{pmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{pmatrix}, \quad B = \begin{pmatrix} 0.6 & 0.5 \\ 0.4 & 0.5 \end{pmatrix}, \quad C = \begin{pmatrix} 1.1 & -0.1 \\ 0 & 1 \end{pmatrix}.$$

- (a) Which matrices are valid transition matrices? Explain each answer.
- (b) A time-homogeneous Markov chain on $\{0, 1\}$ has transition matrix A , in state order $(0, 1)$. Find $\mathbb{P}(X_2 = 1 \mid X_1 = 0, X_0 = 1)$, assuming the conditioning event has positive probability.
- (c) Does the Markov property mean that consecutive states are independent? Explain briefly.

Question 3. A weather model

Tomorrow's weather depends on the past only through today's weather. After a rainy day, tomorrow is rainy with probability $3/4$. After a dry day, tomorrow is rainy with probability $1/4$. Let X_n denote the weather on day n , with states R and D .

- (a) Write the transition matrix P in state order (R, D) .
- (b) Calculate P^2 .
- (c) Given $X_0 = R$, find both $\mathbb{P}(X_2 = R \mid X_0 = R)$ and $\mathbb{P}(X_1 = D, X_2 = R \mid X_0 = R)$.

Question 4. An initial distribution

A Markov chain on $\{0, 1, 2\}$ has transition matrix and initial distribution

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1/2 & 1/2 \end{pmatrix}, \quad \mu_0 = (1/2, 1/2, 0).$$

Find the distributions μ_1 and μ_2 , and then calculate $\mathbb{E}[X_2]$.

Question 5. An urn with two balls

An urn always contains two balls, each red or blue. At each step, choose one ball uniformly and replace it with a ball of the opposite colour. Choices at successive steps use independent randomization. Let X_n be the number of red balls after n replacements. Initially both balls are red.

- (a) Write the state space, the initial state, and the transition matrix in state order $(0, 1, 2)$.
- (b) Find the distribution of X_2 .
- (c) Find the probability that the third ball selected is red. State which X_n describes the urn just before that selection.

Question 6. Two consecutive heads

A fair coin is flipped independently. Let N be the number of flips up to and including the first occurrence of two consecutive heads. Use states 0, 1, 2: before success, states 0 and 1 record the number of trailing heads; state 2 records that two consecutive heads have appeared at least once and is absorbing. Start in state 0.

- (a) Write the transition matrix in state order $(0, 1, 2)$.
- (b) Find $\mathbb{P}(N \leq 4)$.
- (c) Find $\mathbb{P}(N = 4)$. You may use matrix multiplication, a recursion, or direct enumeration of coin sequences.

Question 7. First-step analysis in a fair game

A gambler starts with i units and wins or loses one unit at each independent play, each with probability $1/2$. The game stops when the fortune reaches 0 or 3. Let h_i be the probability of reaching 3 before 0, starting with i units.

- (a) State h_0 and h_3 .
- (b) Condition on the first play to write equations for h_1 and h_2 .
- (c) Solve these equations. Starting with one unit, what is the probability of ruin?

Question 8. A biased game

A gambler starts with one unit. At each independent play, the gambler wins one unit with probability $2/3$ and loses one unit with probability $1/3$. The game stops at fortune 0 or 4.

- (a) Find the probability of reaching 4 before 0, and the probability of ruin. You may use the gambler's ruin formula from Lecture 1.
- (b) If the win probability were instead $1/2$, what would the success probability be? Compare the two success probabilities.

Question 9. Communicating classes

A Markov chain on $\{0, 1, 2, 3\}$ has transition matrix, in that order,

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/4 & 3/4 & 0 & 0 \\ 1/4 & 0 & 1/2 & 1/4 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

- List all communicating classes and identify which are closed.
- Classify every state as recurrent or transient, with a brief reason.
- Can state 1 be reached from state 2, even though $p_{21} = 0$? Give a positive-probability path if one exists.

Question 10. First return versus a visit at a fixed time

Use the chain in Question 9 and suppose $X_0 = 2$. Define

$$T_2^+ = \min\{n \geq 1 : X_n = 2\},$$

with $T_2^+ = \infty$ if there is no positive-time return.

- Find $\mathbb{P}(T_2^+ = 1 \mid X_0 = 2)$ and $\mathbb{P}(T_2^+ < \infty \mid X_0 = 2)$.
- Find $\mathbb{P}(T_2^+ = 2 \mid X_0 = 2)$ and $\mathbb{P}(X_2 = 2 \mid X_0 = 2)$. Explain why these probabilities differ.

Question 11. An unrestricted random walk

A random walk starts at 0 on $\mathbb{Z}$. Its independent increments are +1 with probability $p = 3/4$ and -1 with probability $q = 1/4$. There are no absorbing boundaries.

- Find $\mathbb{P}(X_2 = 0 \mid X_0 = 0)$.
- Find the probability f_0 of ever returning to 0 after time zero, and classify state 0 as recurrent or transient. You may use the Lecture 2 result $f_0 = 2 \min\{p, q\}$.
- If $p = q = 1/2$, is state 0 transient, positive recurrent, or null recurrent? State whether its mean positive return time is finite. No proof is required.

Question 12. A stationary distribution

A machine has two states, working (W) and faulty (F). Its daily transition matrix in state order (W, F) is

$$P = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix}.$$

- Solve $\pi P = \pi$ together with $\pi_W + \pi_F = 1$.
- Interpret π_F as a long-run proportion of days.
- Starting in state W , find the expected number of days until the next visit to W . A visit on the next day counts as a return.

Question 13. A three-state chain

Consider the transition matrix in Question 4.

- (a) Explain why the chain is irreducible and aperiodic.
- (b) Find its stationary distribution by solving the balance equations and the normalization equation.
- (c) Find $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = 2 \mid X_0 = 0)$. State why the convergence theorem from Lecture 2 applies.

Question 14. Stationarity and periodicity

A Markov chain on $\{0, 1\}$ has transition matrix

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

- (a) Find the period of each state and a stationary distribution π .
- (b) Suppose $X_0 = 0$. Find $\mathbb{P}(X_n = 0 \mid X_0 = 0)$ separately for even and odd n . Does this probability have a limit?
- (c) If instead X_0 has distribution π , find $\mathbb{P}(X_n = 0)$ for every $n \geq 0$.

Question 15. Long-run average cost

Use the machine model in Question 12. The cost for a day is 20 units when the machine is working and 80 units when it is faulty.

- (a) Find the long-run average cost per day.
- (b) Given that the machine is working today, find tomorrow's expected cost.
- (c) If the daily cost in state F increases from 80 to 110, while the transition matrix stays unchanged, by how much does the long-run average daily cost increase?

Question 16. Short conceptual questions

Decide whether each statement is true or false. Give a brief justification or a counterexample. You may use examples from the lectures or this assignment.

- (a) Every state of a finite irreducible Markov chain is positive recurrent.
- (b) If a Markov chain starts in a stationary distribution, then its states at consecutive times are independent.
- (c) Every recurrent state has a finite mean positive return time.